

Air Quality Monitoring Node - Bill of Materials with Links

Complete Node Cost Estimate

A single solar-powered air quality sensor node using mid-range components totals approximately **\$61-70**:

Component	Recommended Option	Unit Cost	Link
MCU	LILYGO T-Beam V1.2 868MHz (ESP32 + LoRa + GPS + OLED)	\$29-32	Alibaba - LILYGO Official
PM Sensor	PMS7003 (Plantower)	\$11.50-12	Alibaba - Beijing Ereach
Environmental	BME280 Module (Temp/Humidity/Pressure)	\$2.50-3	Alibaba - BME280 Module
Solar Panel	5V 2W USB Solar Panel	\$3-5	Alibaba - 5V 2W Solar
Battery	Samsung INR18650-30Q 3000mAh	\$2.90-3.10	Alibaba - Samsung 30Q
Charging	TP4056 Type-C + protection	\$0.20-0.50	Alibaba - TP4056 Module
Enclosure	SZOMK IP67 Waterproof Box (130×90×25mm)	\$3-5	Alibaba - SZOMK IoT Enclosure
Cable glands, wiring	PG7/PG9 glands + misc	\$3-5	Alibaba - Cable Glands
Node Total		~\$56-65	

Gateway

Gateway	Type	Channels	Price	Link
RAK7268 WisGate Edge Lite 2	Indoor	8-ch SX1302	\$139-180	AliExpress - RAK Official